Vinit Vaidya

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SUMMARY

MSc Computer Science graduate with 2 years of professional experience in data analytics, machine learning and building automation workflows, complemented by extensive hands-on experience building AI/ML, Generative AI, and data engineering projects. Proficient in Python, SQL, TensorFlow, PyTorch, AWS, Azure, and modern LLM frameworks. Developed enterprise dashboards supporting 160+ stakeholders, production-ready RAG applications, cloud-native MLOps pipelines. Passionate about building scalable AI systems that solve real-world business problems.

WORK EXPERIENCE

Data Analyst | Capgemini **Aug 2021 – Jul 2023**

- Developed **K-Means clustering** (silhouette score 0.70) and **Random Forest classifier** (80% F1-score) for enterprise ticket management, improving routing efficiency and reducing resolution time through quantitative analysis.
- Developed 5 **Tableau** dashboards and optimized 3 existing ones, supporting 160+ stakeholders with data-driven insights that increased engagement by 25%.
- Built automated solutions using **UiPath** with **C#** and **VB.NET** to streamline data extraction, preprocessing, and reporting pipelines, improving efficiency and saving 150 hours annually.
- Developed 5 proof-of-concept for existing applications for integrating **GenAI** with **OpenAI models**, fine-tuned hyperparameters to generate solutions for core business challenges.

PROFESSIONAL DEVELOPMENT

Independent AI/ML Projects & Continuous Learning **Dec 2024 - Present**

- Built production-ready AI applications using Retrieval-Augmented Generation (RAG), LangChain, Amazon Bedrock, FAISS, and Docker.
- Developed cloud-native MLOps pipelines with CI/CD using TensorFlow, Docker, GitHub Actions, AWS, Azure, and DVC.
- Built enterprise-scale data engineering solutions using Azure Databricks, PySpark, Delta Lake, and Medallion Architecture.
- Currently preparing for the AWS Certified Machine Learning – Associate certification while expanding expertise in AI Agents and Generative AI.

KEY PROJECTS

Credit Card Fraud Detection using XGBoost & MLOps Pipeline - *Github Link* **Jul 2026 – Aug 2026**

- Built an end-to-end fraud detection system with an XGBoost classifier on 284K+ imbalanced transactions (0.17% fraud rate), applying SMOTE for class balancing and achieving 0.847 recall, 0.79 precision, and 0.983 ROC-AUC on held-out test data.
- Engineered a config-driven MLOps pipeline with automated data ingestion, schema validation, and MLflow experiment tracking (via DagsHub), and deployed a FastAPI inference service through a Dockerized CI/CD pipeline (GitHub Actions → AWS ECR/EC2).

Chatbot using Retrieval-Augmented Generation - *Github Link* **Sep 2025 – Nov 2025**

- Built an end-to-end RAG solution using Meta Llama3-8B-Instruct for contextual PDF interaction, applying Amazon Titan Embeddings, NLP, and FAISS to generate context-aware responses with 0.80 embedding similarity and 100% self-recall.
- Engineered a dual Dockerized application architecture for document preprocessing, retrieval, and automated deployment, enabling efficient and scalable RAG workflows.

FungiFusion: Integrating Ensemble Deep Learning for Fungi Species Recognition - *Github Link* **Jan 2024 – Aug 2024**

- Designed and trained ensemble deep learning models achieving 85.91% precision/recall through rigorous statistical validation.
- Applied ensemble techniques like bagging, boosting, stacking, and classification to improve quantitative model performance and robustness.

SKILLS SUMMARY

- **Programming & Data:** Python, SQL, R, Tableau, Power BI, NumPy, Pandas, FastAPI
- **Data Science & Machine Learning:** Regression, Classification, Clustering, Feature Engineering, Data Visualization, Statistical Analysis, Scikit-Learn, Ensemble Learning
- **ML Platforms & MLOps (Foundational):** Databricks, MLflow, Model Versioning, Model Monitoring, Experiment Tracking
- **Deep Learning & Advanced Analytics:** TensorFlow, PyTorch, Keras, LangChain, LangGraph (Foundational), Ollama
- **Cloud & Tools:** AWS, Microsoft Azure, Docker, Git, Visual Studio Code, Claude Code, GitHub Copilot

EDUCATION

University of Galway **Sep 2023 - Nov 2024**
MSc. Computer Science GPA: 4.0

Mumbai University **Aug 2017 - Jun 2021**
B.E in Computer Engineering GPA: 3.3